1. What is the clear problem?

Problem: Existing AI-based wind-power forecasting models can provide accurate short-term predictions, but the reliability and stability of their explanations across different wind operating conditions remain insufficiently investigated. This creates a gap in understanding whether XAI methods consistently identify the factors influencing wind-power predictions when the underlying temporal and meteorological conditions change.

Target: Short-term wind-power forecasting
Data: Turbine SCADA + meteorological variables
Prediction: Next 10-minute or 30-minute power output
Models: XGBoost + LSTM
XAI: SHAP
Main investigation: Explanation stability across low/medium/high wind regimes
Evaluation: Forecasting accuracy + explanation stability/reliability

Historical Wind/SCADA Data

↓

Data Preprocessing

↓

Short-Term Power Forecasting

↓

AI Model

(e.g., XGBoost / LSTM)

↓

Predicted Wind Power

↓

XAI

(e.g., SHAP)

↓

"Why did the model make this prediction?"

↓

Explanation Stability Analysis

↓

"Does the explanation remain consistent

under different wind conditions?"

2. What is the purpose?

The purpose of this study is to develop and evaluate an AI-based framework for short-term wind-power forecasting and investigate whether XAI methods can provide stable and reliable explanations of the model's predictions under different wind conditions.

Two connected purposes:

1. Prediction purpose — "Can we predict wind power accurately?"

Use AI models to forecast short-term wind-power generation from historical turbine and weather data.

2. Explanation purpose — "Can we understand and trust the prediction?"

Use XAI to identify which factors influence the prediction and determine whether those explanations remain consistent when wind conditions change.

Expected outcome:
A framework that provides both:
Accurate prediction + understandable and trustworthy explanation

3. What will be the research used concepts?

Concept	Role in your research	Example
Wind Power Forecasting	Main application/problem	Predict power output for the next 10–30 minutes
Time-Series Data	Nature of the data	Historical wind speed, power, rotor speed, temperature
Artificial Intelligence / Machine Learning	Prediction mechanism	XGBoost, LSTM
Explainable AI (XAI)	Explain the predictions	SHAP
Explanation Stability & Reliability	Main research investigation	Check whether explanations remain consistent under different wind conditions

1. Wind Power Forecasting

   This is the **main application domain**.

   The model receives historical information and predicts future turbine power:

   Past observations → AI model → Future wind-power output

   You should specifically define it as **short-term wind-power forecasting**, rather than general wind-energy prediction.

   For example:

   Predict the power generated by a turbine **10 minutes ahead**.

2. Multivariate Time-Series Data
   Wind-power data is not just a collection of independent observations. It has a temporal structure.

   Possible variables include:

   - Wind speed

- Wind direction
- Temperature
- Air pressure
- Rotor speed
- Generator speed
- Pitch angle
- Previous power output

This concept is important because your AI model must learn relationships between variables and their changes over time.

3. Artificial Intelligence / Machine Learning
Baseline: Random Forest or Linear Regression
Main ML model: XGBoost
Deep-learning model: LSTM

This gives you a meaningful comparison between:

Traditional baseline → Tree-based ML → Deep learning

Evaluation can use **MAE, RMSE, R^2**, and possibly MAPE where appropriate.

3. Explainable AI (XAI)

XAI is the second major component.

After the forecasting model produces:

Predicted power = 760 kW

you want to answer:

Why did the model predict 760 kW?

For your first study, **SHAP (SHapley Additive exPlanations)** is a practical primary XAI technique because it can provide both overall feature importance and individual prediction explanations.

4. Explanation Stability and Reliability
Rather than stopping after producing SHAP plots, you investigate whether the explanations are consistent when conditions change.

Low-wind conditions

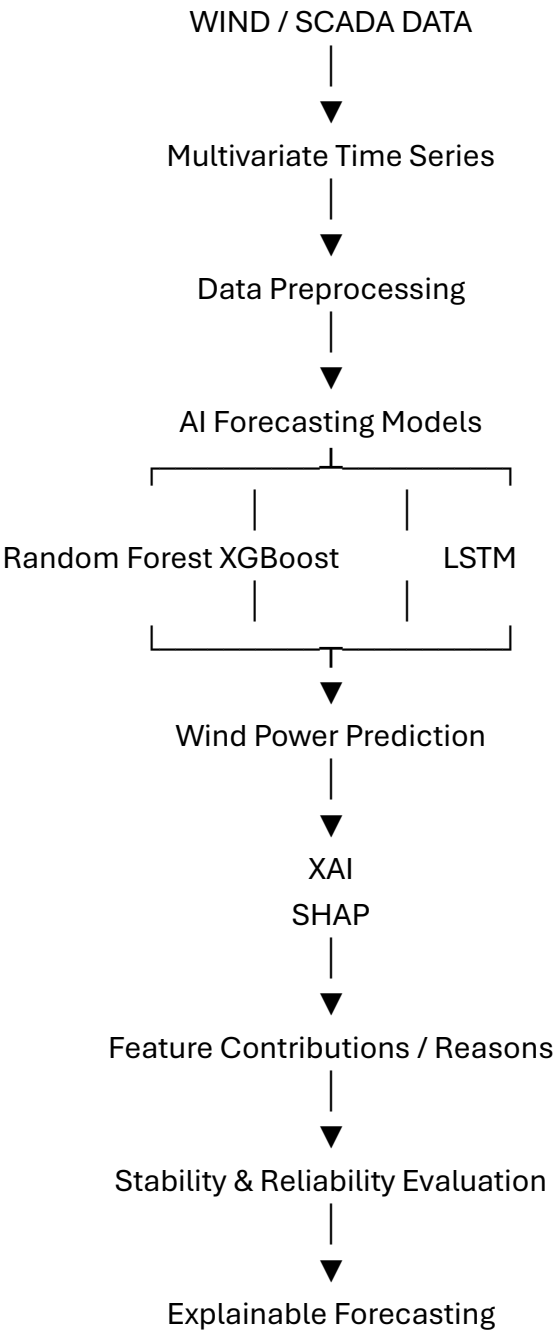
↓

Normal-wind conditions

↓

High-wind conditions

overall conceptual framework



4. What are the research objectives?

Main objective :

To develop and evaluate an AI-based framework for short-term wind-power forecasting and investigate the stability and reliability of XAI explanations under different wind conditions.

Specific objectives :

Objective 1 — Prepare the forecasting dataset

To preprocess and transform multivariate wind-turbine SCADA and meteorological time-series data into a suitable dataset for short-term wind-power forecasting.

This includes handling missing values, outliers, normalization where appropriate, temporal features, and selecting relevant input variables.

Objective 2 — Develop AI forecasting models

To develop and compare AI-based models for short-term wind-power forecasting using historical meteorological and turbine operational data.

compare a baseline model, XGBoost, and LSTM.

Objective 3 — Develop explainability

To apply XAI techniques to identify and quantify the contribution of input features to individual and overall wind-power predictions.

Objective 4 — Evaluate explanation stability

To investigate the stability of XAI explanations across different wind conditions and temporal periods.

Objective 5 — Evaluate the overall framework

To evaluate the forecasting accuracy and the reliability of the generated explanations using appropriate quantitative metrics.

Data → Prediction → Explanation → Stability → Evaluation

5. What are the data collection methods and datasets?

Use the secondary data as the main data-collection method, rather than installing sensors.

6. What are the appropriate approach and design?

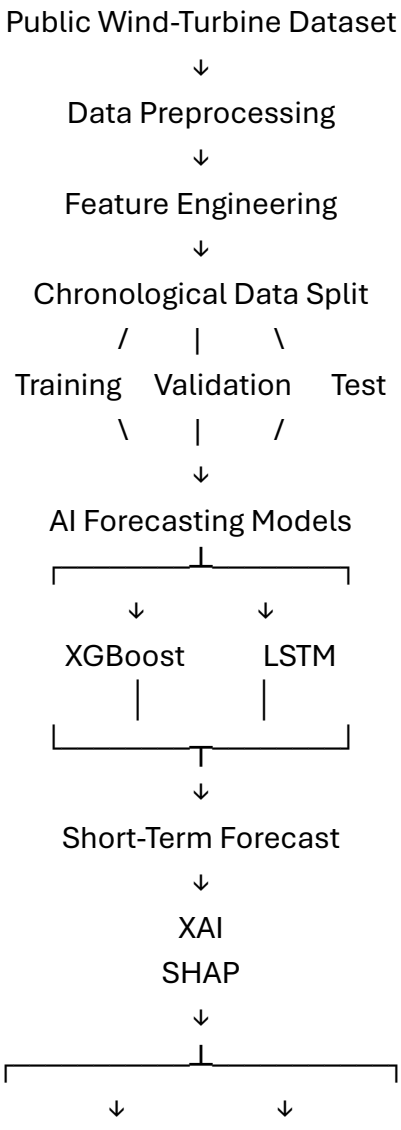
Quantitative empirical research

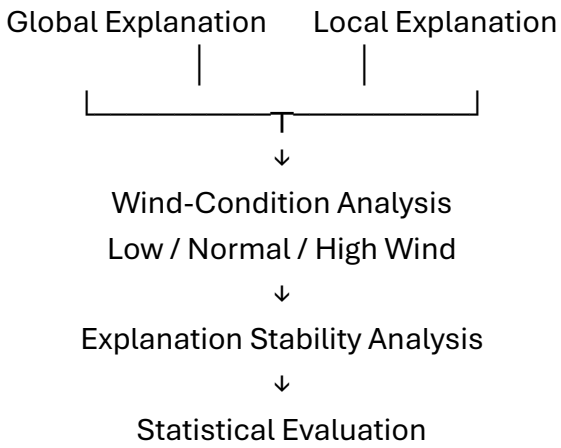
Your research will generate measurable numerical results:

- Wind-power forecasting values
- MAE
- RMSE
- R^2
- Feature importance
- SHAP values
- Explanation stability scores
- Explanation changes across wind conditions

Therefore, the primary approach is:

Quantitative research





1. Data design

Use **one primary wind farm** for the main experiment.

For example:

Multiple turbines from the same wind farm → 10-minute SCADA data → short-term prediction.

Use a **chronological split**, for example:

70% training → 15% validation → 15% testing

The exact percentages can be adjusted depending on the selected dataset.

More importantly, the test data must occur **after** the training data chronologically. This avoids giving the model information from the future.

For stronger validation, you could additionally use an **expanding/rolling-window evaluation**, which is particularly appropriate for time-series forecasting. Recent wind-power research has used temporal-domain and expanding-window approaches because wind-power data can experience distribution shifts over time.

2. AI model design

Model	Purpose
Linear Regression	Simple baseline
Random Forest	Traditional ML benchmark
XGBoost	Strong tree-based ML model
LSTM	Deep-learning/time-series model

Your objective is not simply to find the model with the highest accuracy. The forecasting models provide the basis for your main investigation: how stable and reliable are their explanations?

3. XAI design

For your initial study, I recommend **SHAP as the primary XAI technique**.
It can provide:

Global explanation
Which variables generally influence wind-power prediction?

and

Local explanation
Why did the model predict this particular power value at this particular time?

4. Your most important experiment: explanation stability

This is where your research becomes more distinctive.
Divide the test observations into wind regimes

Low wind
↓
Normal wind
↓
High wind

Then calculate SHAP explanations separately.
You can investigate:

- Is wind speed the dominant feature in all conditions?
- Do the top five important features remain similar?
- Does the feature ranking change considerably between conditions?
- Does the model's explanation remain consistent across different time periods?

You can quantify this using measures such as:

Spearman rank correlation
Compare feature rankings between different periods/regimes.

Top-k feature overlap / Jaccard similarity
Measure how many of the most important features are shared.

SHAP variability
Measure how much feature contributions vary across windows or conditions.

This is especially relevant because recent research has explicitly investigated feature stability in wind-power forecasting, meaning your study needs to define a precise stability evaluation rather than simply claiming that SHAP makes the model trustworthy.

5. **Experimental comparison**
   Your experiments could be structured as follows:

Experiment	Question
Experiment 1	How accurately can the AI models forecast short-term wind power?
Experiment 2	Which input variables influence the forecasts?
Experiment 3	Do explanations differ between low, normal and high wind conditions?
Experiment 4	How stable are explanations across different temporal periods?
Experiment 5	Does the best forecasting model also provide stable explanations?

Overall Methodology

Quantitative empirical research using an experimental and comparative design. The study will use publicly available wind-turbine SCADA time-series data to develop and compare AI models for short-term wind-power forecasting. Chronological training, validation, and testing will be employed to preserve the temporal structure of the data. SHAP will be used to generate global and local explanations, and explanation stability will be quantitatively evaluated across different wind conditions and temporal periods. Forecasting performance and explanation quality will then be compared to assess the effectiveness and reliability of the proposed explainable forecasting framework.

Final Structure

- **Research paradigm**: Quantitative
- **Research approach**: Empirical
- **Research design**: Experimental + comparative
- **Data**: Secondary time-series SCADA data
- **Validation**: Chronological/rolling temporal validation
- **AI**: XGBoost + LSTM + baseline
- **XAI**: SHAP
- **Main investigation**: Explanation stability and reliability
- **Evaluation**: Forecast accuracy + explanation stability

7. What are the limitations?

a. Dependence on publicly available datasets
   The research will primarily use **secondary SCADA and meteorological datasets**. The availability, quality, sampling frequency, and completeness of these datasets are controlled by the original data providers.
   Some datasets may contain:
   - Missing observations
   - Sensor errors
   - Outliers
   - Inconsistent timestamps
   - Limited variables
   - Periods of turbine downtime

   Therefore, the findings may depend partly on the characteristics of the selected dataset.

b. Limited generalizability across wind farms
   A model developed using data from one wind farm may not perform equally well at another location.
   Wind-power generation depends on factors such as:

   - terrain,
   - wind climate,
   - turbine technology,
   - turbine capacity,
   - operating conditions.

   Therefore, if your main experiment uses one wind farm, you should **not claim that the results apply to all wind farms**.

   A second wind farm can be used for external validation to reduce this limitation.

c. XAI explanations are not necessarily ground truth
   This is one of the **most important limitations** in your research.
   SHAP can explain how the trained model associates features with a prediction, but an XAI explanation does not automatically prove that the identified feature is a true physical cause of the wind-power change.
   For example:
   SHAP says wind speed contributed strongly to a prediction.
   That does not necessarily mean:
   wind speed alone physically caused that change in power.
   Your research should therefore describe SHAP explanations as model-based explanations, not definitive physical causation.

d. Explanation stability depends on the XAI method

Your study may use SHAP as the primary XAI technique. Different XAI methods can produce different explanations.

For example:

SHAP → one feature ranking
LIME → another feature ranking

Therefore, conclusions about explanation stability may be specific to the selected XAI method.

A future study could compare multiple XAI techniques.

e. Difficulty in defining "stable" explanations

This is actually a methodological limitation that you should acknowledge.

There is no universally applicable single threshold for saying:

"This explanation is stable."

You therefore need to **define an operational measure of stability**.

For example:

- Feature-ranking correlation
- Top-k feature overlap
- SHAP-value variation
- Explanation consistency across time windows

Your choice of stability metric can influence the final conclusions.

f. Temporal distribution changes

Wind conditions and turbine behavior can change over time.

Training period
↓
Different wind/weather patterns
↓
Testing period
↓
Different data distribution

A model that performs well during one period may perform differently later.

This is particularly important for wind-power forecasting because the data-generating environment is not completely stationary. Recent research has examined temporal distribution shifts and feature stability in wind-power forecasting.

This is one reason your chronological or rolling-window evaluation is preferable to a simple random split.

g. SCADA data may not contain all relevant physical variables
Even a rich SCADA dataset may not include important factors such as:
- Turbulence characteristics
- Local terrain effects
- Wake effects between turbines
- Detailed weather observations
- Maintenance conditions
- Grid curtailment
- Certain turbine-control parameters

Consequently, the AI model may be unable to fully explain some variations in power output.

h. Computational limitations
Deep-learning models such as LSTM can require considerable computational resources, particularly with:
- Large SCADA datasets
- Multiple turbines
- Long historical sequences
- Hyperparameter optimization
- Extensive XAI calculations

This may limit the number of models, turbines, or experimental configurations that can reasonably be evaluated.

i. Forecast horizon limitation
Your proposed study focuses on **short-term forecasting**, such as 10-minute or 30-minute prediction.
Therefore, the findings should not automatically be generalized to:
- Day-ahead forecasting
- Week-ahead forecasting
- Long-term wind-energy planning

Different forecasting horizons can require different models and features.

j. Accuracy versus interpretability trade-off

A more complex AI model may provide better predictive performance but may also be more difficult to interpret.

For example:

LSTM/Transformer → potentially strong predictive capability
Simpler models → potentially easier interpretation

Your study therefore needs to consider **both forecasting performance and explanation quality**, rather than selecting a model solely because it has the lowest RMSE.

Limitations of the Study

This study has several limitations. First, the research will primarily rely on publicly available wind-turbine SCADA and meteorological datasets. Consequently, the availability, quality, temporal resolution, and completeness of the data will depend on the characteristics and documentation provided by the original data sources.

Second, the main analysis will be conducted using data from a selected wind farm or a limited number of wind farms. Therefore, the findings may not be directly generalizable to all geographical regions, turbine technologies, or operating environments. Where feasible, an independent dataset from another wind farm may be used to assess the generalizability of the proposed framework.

Third, the XAI methods used in this study, particularly SHAP, provide explanations of the relationships learned by the AI model and should not be interpreted as definitive evidence of physical causality. Different XAI techniques may also produce different explanations for the same prediction.

Fourth, explanation stability and reliability must be defined through measurable criteria such as feature-ranking consistency, feature overlap, or variation in explanation values. The selected measures and thresholds may influence the assessment of explanation stability.

Fifth, wind-power generation is influenced by changing meteorological and operational conditions. Temporal changes and distribution shifts between training and testing periods may affect both forecasting performance and XAI explanations. Although chronological and/or rolling-window validation will be used to reduce this issue, such changes cannot be completely eliminated.

Finally, the study will focus on short-term wind-power forecasting and therefore its findings may not be directly applicable to longer-term forecasting problems. In addition, computational resources may constrain the scale of deep-learning models, hyperparameter optimization, and extensive XAI experimentation.

8. What are the analysis and interpretations?

1. Exploratory Data Analysis

You would analyze:

- Distribution of wind speed
- Distribution of power output
- Relationship between wind speed and power
- Correlation between variables
- Missing values
- Outliers
- Temporal patterns
- Changes in power output over time
- Low-, normal-, and high-wind operating conditions

Wind Speed

↓

Power Output

↓

Does power increase as wind speed increases?

↓

Are there unusual observations?

↓

Are relationships different under different conditions?

2. AI Forecasting Performance Analysis

Model	MAE	RMSE	R^2	Training Time
Linear Regression	—	—	—	—
Random Forest	—	—	—	—
XGBoost	—	—	—	—
LSTM	—	—	—	—

You should also examine performance separately under:

Low wind → Normal wind → High wind

Interpretation example

Suppose:

XGBoost: RMSE = 85 kW
LSTM: RMSE = 73 kW

You would not simply say:

“LSTM is better.”

Instead:

“LSTM produced a lower RMSE than XGBoost, indicating that it captured the temporal patterns in the wind-power series more effectively for the selected forecasting horizon.”

3. XAI Analysis

Global explanation

1. Wind Speed [redacted]
2. Previous Power [redacted]
3. Rotor Speed [redacted]
4. Wind Direction [redacted]
5. Temperature [redacted]

Local explanation

Wind speed +120 kW

Previous power +45 kW

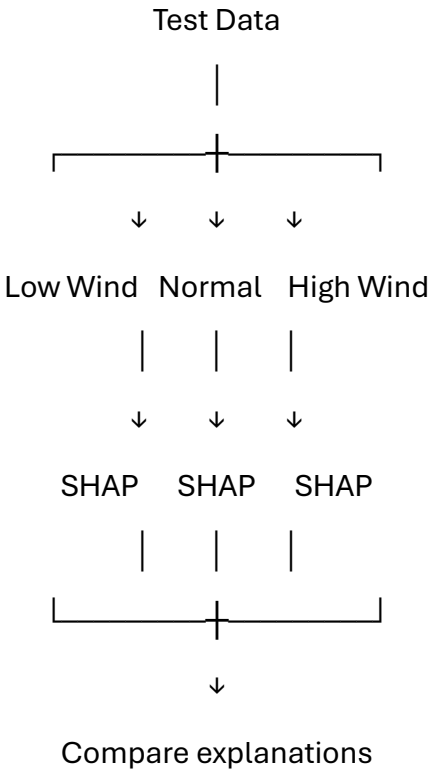
Rotor speed +20 kW

Temperature -5 kW

Wind direction -12 kW

Explanation Stability Analysis

This should be the **core analysis of your research**.



You can analyze:

Feature-ranking stability

Does the order of important variables remain similar?

Top-k overlap

Are the same important variables repeatedly identified?

SHAP-value variation

How much do individual feature contributions change?

Temporal stability

Do explanations remain consistent across different months, seasons, or test windows?

Feature	Low Wind Rank	Normal Wind Rank	High Wind Rank
Wind speed	**1**	**1**	**1**
Previous power	**2**	**2**	**3**
Rotor speed	**4**	**3**	**2**
Wind direction	**3**	**4**	**5**

Then you can quantitatively calculate consistency.

This is particularly important because recent research has moved toward quantitative trustworthiness and feature-stability evaluation rather than treating an XAI visualization as inherently trustworthy.

Relationship Between Accuracy and Explanation Stability

Model	Forecast Accuracy	Explanation Stability
XGBoost	High	High
LSTM	Very High	Moderate
Random Forest	Moderate	High

6. Statistical Analysis

Model performance

+

XAI importance

+

Explanation stability

+

Statistical testing

↓

Evidence-based conclusion

7. **How to interpret the final results**

Question 1 — Which AI model predicts best?
Interpret MAE, RMSE and R^2.

Question 2 — Which variables influence predictions?
Interpret SHAP global and local results.

Question 3 — Does feature importance change with wind conditions?
Compare low/normal/high regimes.

Question 4 — Are XAI explanations stable?
Use your selected stability metrics.

Question 5 — Is there a relationship between prediction quality and explanation quality?
Compare the forecasting and XAI results together.

Complete Analysis Framework

Dataset and Exploratory Analysis

↓

Data Preprocessing Analysis

↓

AI Forecasting Performance

↓

Forecasting Performance by Wind Regime

↓

Global XAI Analysis

↓

Local XAI Analysis

↓

Explanation Stability Analysis

↓

Temporal Stability Analysis

↓

Statistical Significance / Robustness Analysis

↓

Accuracy vs. Explanation Stability

↓

Overall Interpretation

Full Research Story Have Developed

- **Problem** → AI forecasts are difficult to interpret, and explanation stability under changing wind conditions is insufficiently established.
- **Purpose** → Develop and evaluate an explainable short-term wind-power forecasting framework.
- **Concepts** → Wind forecasting + time-series data + AI + XAI + explanation stability/reliability.
- **Objectives** → Prepare data → develop forecasting models → explain predictions → evaluate explanation stability → evaluate overall performance.
- **Data** → Public SCADA/time-series wind-turbine data, preferably one wind farm for the main experiment.
- **Approach/design** → Quantitative empirical + experimental/comparative + chronological time-series validation.
- **Limitations** → Dataset dependence, site generalizability, XAI/model dependence, stability-definition issues, temporal changes, and scope limitations.
- **Analysis** → Forecast accuracy + wind-regime analysis + SHAP + stability/reliability + statistical comparison.

9. What are the suggested titles?

Best overall

"Explainable AI for Short-Term Wind Power Forecasting: Evaluating Explanation Stability across Wind Regimes and Temporal Periods"

More academic

"Assessing the Stability and Reliability of Explainable AI in Short-Term Wind Power Forecasting"

More methodology-oriented

"An Explainable AI Framework for Short-Term Wind Power Forecasting with Stability Analysis of Feature Attributions"

10. Research Gap

Recent research has demonstrated the value of Explainable Artificial Intelligence (XAI) in wind-power forecasting by using methods such as SHAP, LIME, permutation feature importance, and partial dependence analysis to identify the factors influencing AI-based forecasts. A 2024 study specifically proposed quantitative metrics for assessing the trustworthiness of XAI interpretations in wind-power forecasting. More recent studies have also applied SHAP to deep-learning-based wind-power forecasting and examined feature contributions to power predictions.

However, recent literature indicates that important challenges remain. A 2025 review of XAI for wind-energy systems identifies limited temporal explainability and the lack of suitable benchmarking datasets as continuing challenges. Furthermore, a 2026 study has explicitly investigated feature stability in SHAP-based wind-power forecasting, demonstrating that explanation stability is now an emerging research direction rather than an entirely unexplored problem.

Therefore, the research gap addressed by this study is not the general absence of XAI or feature-stability research. Instead, there is a need for a focused and reproducible investigation of **how the explanations of short-term wind-power forecasting models vary across different wind regimes and temporal periods, and whether a model with better forecasting accuracy also provides consistently stable explanations**.

In particular, this study will investigate explanation stability under low-, normal-, and high-wind conditions and across chronological test periods, while comparing selected AI forecasting models. The study will connect forecasting performance with explanation stability, thereby providing a more comprehensive assessment of AI models than accuracy-based evaluation or qualitative XAI visualization alone.

The intended contribution is a systematic experimental framework for evaluating the relationship between **short-term forecasting accuracy, feature attribution, and explanation stability** in wind-power forecasting.

11. Problem Statement

Short-term wind-power forecasting is challenging because wind conditions and turbine operating states vary over time, resulting in complex and nonlinear relationships between meteorological and operational variables and generated power. AI-based forecasting models can learn these relationships and produce accurate predictions, but their decision-making processes are not directly transparent to users.

XAI methods such as SHAP can provide information about the features contributing to individual and overall predictions. However, the existence of an explanation does

not by itself establish that the explanation is stable or reliable under changing operating conditions. Although recent studies have begun to examine trustworthiness and feature stability in XAI-based wind-power forecasting, temporal explainability remains an identified challenge, and the relationship between forecasting performance and explanation stability requires further systematic investigation.

Consequently, the problem addressed in this study is the insufficiently characterized stability of XAI explanations for short-term wind-power forecasting when wind conditions and temporal conditions change. It is therefore necessary to determine whether the important features identified by XAI remain consistent across different wind regimes and time periods, and whether the forecasting model that provides the best predictive performance also provides sufficiently stable explanations.

This study will address this problem by developing selected AI-based short-term forecasting models, applying SHAP to explain their predictions, and quantitatively evaluating explanation stability across different wind conditions and chronological periods alongside conventional forecasting-performance metrics.

12. Research Questions

RQ1. How accurately can selected AI models forecast short-term wind-power generation using multivariate wind-turbine SCADA and meteorological time-series data?

RQ1 → Prediction
Which AI model forecasts best?

RQ2. Which meteorological and turbine-operational features have the greatest influence on short-term wind-power predictions made by the selected AI models?

RQ2 → Explanation
Why does the model make its predictions?

RQ3. How do XAI feature attributions change across different wind conditions, such as low-, normal-, and high-wind regimes?

RQ3 → Condition dependence
Do explanations change when wind conditions change?

RQ4. How stable are XAI explanations across different chronological periods and changing wind conditions?

RQ4 → Stability
Are those explanations consistent over time?

RQ5. What is the relationship between forecasting performance and XAI explanation stability, and does the most accurate forecasting model also provide the most stable explanations?

RQ5 → Overall insight
Does higher prediction accuracy necessarily mean better/stabler explanations?

Research idea → **research gap** → **research questions** → **objectives** → **dataset** → **feature engineering** → **AI/ML models** → **methodology** → **experiments** → **evaluation metrics** → **explainable AI** → **novelty** → **paper structure** → **suitable journals/conferences**.
